Hackathon Technical Solution Report

Field	Details
Problem ID	P1
Problem Title	Robust Low-Light Image Restoration and Enhancement
Team Name	APex decs
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GitHub Repository	
Live Demo / Deployment	

1. Problem Understanding & Solution Strategy

- **Core Technical Challenge:** Photos taken in the dark have four problems at once: they are very underexposed, they are full of sensor noise that gets bigger when the image is brightened, they often have a colour cast, and the light is uneven (a lamp in one corner, deep shadow in another). Simply raising brightness makes the noise worse and turns lamps pure white. Our constraints were to run on a normal laptop CPU with no GPU, answer in under about half a second per 600x400 image, and show every step in a web app.
- **Proposed Methodology:** We run a classical Retinex method and a small deep learning model (Zero-DCE) side by side, then mix them pixel by pixel using the estimated illumination map: the darkest regions take the Retinex result and the lit regions take Zero-DCE. Every strength (brightening target, denoising, CLAHE, white balance) is computed from measurements of the input image, and the two fusion settings were chosen on a validation set. We chose this because neither method alone works everywhere: on the LOL-v2 Real test set the hybrid scores 19.63 dB, against 18.02 dB (classical) and 18.12 dB (Zero-DCE).

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2. Technical Architecture & Implementation

The pipeline diagram is in `docs/architecture.png`. Code: `src/classical.py`, `src/zerodce.py`, `src/hybrid.py`, metrics in `src/metrics.py`, app in `app.py`.

2.1 Classical Algorithmic Baseline

- **Algorithms Utilized:**
  1. **Illumination map T** (LIME idea): the maximum of R, G, B for every pixel, smoothed with a **guided filter** that uses the image itself as the guide, so edges between bright and dark objects stay sharp and no halos appear.

2. **Retinex enhancement:** image = reflectance x illumination, so we change only the light: output = I x T^(gamma - 1). A floor of 0.02 on T stops black pixels exploding.
3. **Bilateral filter** in YCrCb, stronger on the two colour channels, because colour noise looks worst.
4. **CLAHE** on the L channel of LAB only, so colours do not shift.
5. **Grey-world white balance** (partial) and a soft knee above 0.85 so bright lights fade to white instead of clipping.

- **Adaptive Selection Logic:** For each input we measure mean brightness, share of very dark pixels, noise level (wavelet estimate), colour cast, contrast (95th minus 5th percentile) and how uneven the light is. From these:
  - gamma is solved so the mean of the new illumination reaches 0.45 (0.38 for extremely dark images, so we do not just reveal noise);
  - denoise strength = noise level x expected brightening gain x 6, clipped to 0..1, so a clean photo is barely smoothed and a dark noisy one is smoothed strongly. Denoising is also blended by T, so the darkest regions (amplified most) get the most smoothing;
  - the CLAHE clip limit is 2.0 for flat images and 1.5 otherwise, reduced further when the image is noisy;
  - white balance is applied only when the colour cast is strong, and at most 30%.

2.2 Deep Learning Pipeline

- **Model Architecture: Zero-DCE** (Guo et al., CVPR 2020), a 7-layer CNN with 79,416 parameters. Instead of an image it predicts 8 per-pixel curves, $x + r(x^2 - x)$, which are smooth and always stay in 0..1, so it cannot invent colours or textures. We chose it over EnlightenGAN or RetinexNet because it is tiny enough for a CPU web app and a GAN can hallucinate detail. Our changes: curves are predicted on a copy of at most 512 px and upsampled; the output after 4 curves and after 8 curves are both computed and the one with brightness closer to 0.5 and fewer clipped pixels is kept; `torch.set_flush_denormal(True)` cut CPU time for one 512 px image from 13.6 s to 0.37 s.
- **Loss Functions & Training:** We started from the authors' weights and fine-tuned on the official **LOL-v2** dataset (Real_captured + Synthetic Train, 1,589 pairs: 1,431 for training, 158 held back for validation, random seed 0). Loss = the four Zero-DCE losses (spatial consistency, exposure control at 0.5, colour constancy, smoothness of the curve maps) + 20 x L1 distance to the ground truth. Adam, learning rate 2e-4, 15 epochs, 128 px crops, batch 8, on a MacBook CPU. Validation PSNR rose from 13.23 to 14.58 dB (best epoch 6).

2.3 Hybrid Integration & Adaptive Mechanism

- **Fusion Strategy:** Classical priors guide the network in two places.
  1. **Pre-gain:** when the image mean brightness is below 0.10, it is first lifted with the Retinex map to a mean of about 0.20 before going into Zero-DCE, because Zero-DCE was trained on images that are dark but not black.

2. **Illumination-guided fusion:** each pixel gets a Zero-DCE weight w = clip(T / 0.2, 0, 1), smoothed with a Gaussian (sigma 5) to avoid seams, and output = w x ZeroDCE + (1 − w) x Retinex. The threshold 0.2 and the CLAHE strength 0.35 were chosen on the 158 validation pairs (`results/tune_fusion_official.txt`). Our first fusion rule (exposure-fusion weights) did worse than a plain average, because Zero-DCE's noise looked like "detail" and attracted weight.

- **Decision Logic:** Zero-DCE gives natural brightness and colour where there is some light, but in the darkest areas it under-lights and amplifies noise. The Retinex branch is steadier there. So the illumination map decides: low T means trust Retinex and denoise hardest; high T means trust Zero-DCE and keep detail. After fusion, denoising strength follows the noise estimate and is spatially weighted by T, CLAHE is applied lightly, and the highlight knee protects bright regions, giving 0% over-exposed pixels on all three test sets.

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3. Experimental Benchmarks & Evaluation

3.1 Quantitative Benchmarks

Official **LOL-v2 Real_captured Test** set (100 pairs, never used for training or tuning). LPIPS on the same 100 pairs; NIQE on 26 real dark photos without ground truth (DICM, LIME, RetinexNet samples). Latency on a MacBook Pro CPU, 600x400 images.

Approach / Model Variant	PSNR (dB) ↑	SSIM ↑	Perceptual Metric (LPIPS / NIQE) ↓	Latency (ms / frame)
Classical Baseline	18.02	0.673	0.286 / 4.55	69 ms
Deep Learning Only	18.12	0.580	0.313 / 4.75	380 ms
Proposed Hybrid System	19.63	0.761	0.230 / 4.76	447 ms

Deep Learning Only = Zero-DCE with the authors' weights (the better of the two Zero-DCE variants on this set; our fine-tuned version scored 15.58 dB / 0.485). Other test sets:

Test set	Classical PSNR / SSIM	Zero-DCE (best) PSNR / SSIM	Hybrid PSNR / SSIM
LOL-v2 Synthetic Test (100)	15.51 / 0.671	16.98 / 0.724	16.95 / 0.722
LOL v1 eval15 (15)	16.37 / 0.656	16.79 / 0.520	18.37 / 0.742

- **Analysis:** The hybrid gives the best structural fidelity (+1.5 dB PSNR, +0.09 SSIM over the best single method) and the best LPIPS on real captured data, at about 6.5x the

classical method's time, since Zero-DCE takes most of the 447 ms. NIQE slightly prefers the classical output (4.55 vs 4.76) because NIQE rewards strong local contrast, while the hybrid is smoother and closer to the reference photos. On LOL-v2 Synthetic, which has little real noise, the hybrid only ties with Zero-DCE.

3.2 Visual Outputs & Intermediate Representations

All figures are made by `python scripts/make_figures.py` and stored in `docs/`.

1. **Extreme low-light:** `docs/compare_lolv2.jpg`, rows 1–2: the two darkest LOL-v2 Real test images, shown as input | classical | Zero-DCE | hybrid | ground truth.
2. **Non-uniform / backlit scenes:** `docs/compare_lolv2.jpg`, row 3 (the LOL-v2 test image with the most uneven light) and `docs/compare_real.jpg` (real night photos with lamps and windows). Bright regions are kept below clipping by the highlight knee: 0% over-exposed pixels for the hybrid against 0.013% for Zero-DCE on LOL-v2 Real.
3. **Intermediate states:** `docs/intermediate.jpg` shows input, illumination map T, Retinex branch, Zero-DCE branch, fusion weight map (bright = Zero-DCE), fused, denoised and final output. The web app shows the same steps for any uploaded photo, together with the measured noise level and the parameters chosen from it.

3.3 Ablation Study

Each part is switched off once and the same 100 LOL-v2 Real test pairs are re-run (`results/ablation_lolv2_real.csv`).

Change to the hybrid	PSNR (dB)	Δ PSNR	SSIM	Δ SSIM
Full hybrid	19.63	–	0.761	–
No classical guidance in fusion (Zero-DCE branch only)	13.91	−5.73	0.618	−0.144
Fusion replaced by a fixed 50/50 average	18.01	−1.62	0.715	−0.046
No denoising	18.04	−1.59	0.497	−0.265
Fixed instead of adaptive parameters	19.33	−0.30	0.741	−0.020
No pre-gain	19.36	−0.27	0.759	−0.002
No colour correction	19.45	−0.18	0.760	−0.001
Authors' Zero-DCE	19.67	+0.03	0.762	+0.001

Change to the hybrid	PSNR (dB)	Δ PSNR	SSIM	Δ SSIM
weights instead of fine-tuned				
No CLAHE	20.29	+0.66	0.784	+0.023

Classical guidance is the single most important part, and illumination-based weighting beats plain averaging by 1.62 dB. Adaptive parameters help and also stop clipping (fixed parameters give 0.006% over-exposed pixels). Removing CLAHE would score higher on the test set, but we did not choose settings from the test set: on the validation set the light CLAHE setting (0.35) scored best (14.72 dB, against 14.44 dB with CLAHE off), so we kept it.

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4. Failure Modes & Limitations

- **Observed Failure Case:** Fine-tuning made Zero-DCE alone *worse* on real captured photos: on LOL-v2 Real it fell from 18.12 to 15.58 dB and its remaining noise rose from 0.032 to 0.056, while it improved on LOL-v2 Synthetic (16.69 → 16.98 dB) and eval15 (14.98 → 16.79 dB). The three test images where the hybrid scores lowest are shown in `docs/failures_lolv2.jpg` (input | classical | Zero-DCE | hybrid | ground truth).
- **Root Cause & Mitigation:** Training and validation mixed 689 Real and 900 Synthetic pairs, so the validation score was mostly synthetic and the saved epoch was the one that brightens aggressively. Real sensor noise is much heavier, and a curve that multiplies brightness multiplies noise by the same factor, with no step in Zero-DCE to remove it. In the hybrid this is mostly hidden because the darkest regions use the Retinex branch and are denoised hardest (the hybrid is equal with either weights: 19.63 vs 19.67 dB). Fixes: validate on Real images only, or keep one model per domain; and replace the bilateral filter with a small learned denoiser for extreme shadows, where bilateral filtering cannot separate fine texture from noise. Other limits: very bright scenes come out too grey (we aim at average brightness), and scenes that are truly one colour can be shifted by the grey-world white balance.

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5. Deployment & Quickstart Guide

Quickstart Commands

```
# 1. Clone the repository and install dependencies
git clone <your-github-repo-url>
cd lowlight-restore
pip install -r requirements.txt

# 2. Run inference on a sample directory
python run_inference.py --input_dir ./samples/ --output_dir ./results/

# optional: classical or DL only, and save the intermediate maps
python run_inference.py --input_dir ./samples/ --output_dir ./out/ --
```

```
method classical
python run_inference.py --input_dir ./samples/ --output_dir ./out/ --
save_steps

# 3. Web app (http://localhost:7860)
python app.py

# 4. Reproduce all benchmark numbers (LOL-v2 in ./LOL-v2, LOL v1
in ./LOLdataset)
sh scripts/run_official.sh LOL-v2 LOLdataset
```

Deployment Format

- **Gradio UI** (`app.py`): upload a photo, choose Hybrid / Classical / DL, see a before/after slider, every intermediate step and the measured image statistics, plus a tab that runs all three methods side by side with timings. Large photos are resized to 1280 px, and there are clear error messages for a missing or too-small image.
- **Docker container** (`Dockerfile`): `docker build -t lowlight . && docker run -p 7860:7860 lowlight`.
- **CLI script** (`run_inference.py`) for folders of images, which also writes `latency.csv`.
- Hosted on Hugging Face Spaces (CPU); see `docs/DEPLOY.md`.

Student Submission Checklist

- ☐ Working GitHub repository link included with an updated `README.md`.
- ☒ No raw code pasted into this report.
- ☒ All table metrics match test runs reproducible via the repository.
- ☐ All team member details accurately recorded.